

Number Shapes

Numbers you can build out of dots

1. Square numbers: draw the next one

These squares of dots are 1, 4, 9, and 16. Count to check.



Now draw the next square number, 5 across and 5 up, in the box:



It has _____ dots.

2. Triangle numbers: draw the next one

These triangles are 1, 3, 6, and 10. Each one adds a longer row.



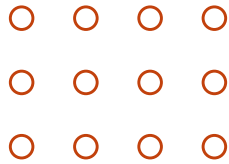
Now draw the next triangle number, with a new bottom row of 5, in the box:



It has _____ dots.

3. Fit two triangles into a rectangle

Build the triangle **6** (rows of 1, 2, 3) twice, in two colors. Turn one upside down and slide them together. Color this grid: make one triangle orange and the other blue.



The rectangle is _____ by _____, which is _____ dots.

One triangle is half of that, so it is _____ dots.

4. Squares grow by odd numbers

Look at the square numbers in order. Write the jump on each line:

1 +_____ 4 +_____ 9 +_____ 16 +_____ 25

What kind of numbers are all the jumps? _____

5. Your own number shape

Pick any number of coins or blocks. Can you make a square with them? A triangle? Draw what you build:

